Title : TikTok Claims Classification Project

Subtitle : Data Inspection and Exploratory Analysis (EDA)

Project Overview

The goal of this project is to develop a machine learning model that streamlines the moderation process by identifying whether statements in videos are **claims** or **opinions**. Claims refer to unsourced or unverified information, while opinions represent personal beliefs. This inspection phase focuses on organizing the data and identifying key variables that drive engagement and claim status.

Key Insights

A Balanced Foundation: The dataset is split nearly evenly, with **50.3% claims** and **49.7% opinions**, providing a stable baseline for model training.
The "Viral" Indicator: Claim-based content is the primary driver of massive reach on the platform.

- **Claims Median Views: 501,555**.
- **Opinions Median Views: 4,953**.

Safety & Ban Correlation: There is a significant correlation between account status and content type. Authors who are **banned** or **under review** are overwhelmingly likely to have posted claims.
Verification Gap: The vast majority of claim-type content originates from **unverified** accounts.

Details

Data Integrity: The dataset contains **19,382 rows** and 12 columns.
Missing Values: Approximately **298 rows** (1.5%) have null values in critical fields like claim_status and engagement metrics. These need addressing to prevent model bias.
Metric Skewness: Engagement metrics (likes, shares, comments) are heavily right-skewed. The average engagement for claims is significantly higher than for opinions across every metric.

Next Steps

Data Cleaning: Remove or impute the 298 missing records to ensure high-quality training data.
Feature Engineering: Further explore the **likes-to-view** and **share-to-view** ratios as these normalized metrics may better distinguish claims from opinions than raw volume alone.
Modeling Strategy: Prioritize non-parametric or tree-based models (e.g., Random Forest) to handle the extreme outliers and skewness found in the engagement data.

Title : Claims Classification: Data Preprocessing Phase

Subtitle : Developing Automated Moderation Signals

ISSUE / PROBLEM

TikTok needs to automate the moderation of videos by distinguishing between **claims** (unverified info) and **opinions** (personal thoughts). Currently, the manual review process is backlogged. We need a data-driven way to prioritize high-risk content for review.

RESPONSE

We analyzed **19,382 videos** to identify patterns that signal a "claim".

- **Normalized Metrics**: Created engagement ratios (likes-to-views) to identify content intensity.
- **Data Audit**: Identified **298 incomplete rows** that must be removed before modeling.
- **Segment Analysis**: Compared engagement across **9,608 claims** and **9,476 opinions**.

IMPACT

Engagement Gap: Claims average **501,000+ views**, while opinions average only **~5,000**. High views are a near-certain indicator of a claim.
Risk Correlation: Banned authors are **8x more likely** to post claims than opinions.
Efficiency: Using these signals, the team can automate the identification of claims with high

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KEY INSIGHTS

Balanced Data: The 50/50 split between claims and opinions is ideal for training a stable model.
Banned Authors: These users have significantly higher median engagement, suggesting their "viral" claims triggered the bans.
Statistical Skew: Data is heavily right-skewed; we must use models like **Random Forest** that handle outliers well.